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A

1 Dock and USB protection

2 Charger, battery, 3.0 V

3 Microcontroller

B

4 Microphone

5 Speaker driver and ear pad

C

HOW TO READ THIS

Each box above is one page. Power arrives on page 1 (the dock), is stored and cleaned on page 2, and feeds the brain on page 3. Sound comes in on page 4 (the microphone) and goes out on page 5 (the speaker driver). Flag-shaped labels with the same name are the same wire, even on different pages. Arrow and bar symbols (+3V0, VSYS, VBAT, GND) are power rails. Yellow boxes explain each block in plain words.

Stereo Ultrasound

Sheet: /

File: pod.kicad_sch

Title: Pod board Rev G: schematic (generated from gen.py)

Size: A4

Date: 2026-10-02

Rev: G

KiCad E.D.A. 10.0.6

Id: 1/6

D

1

2

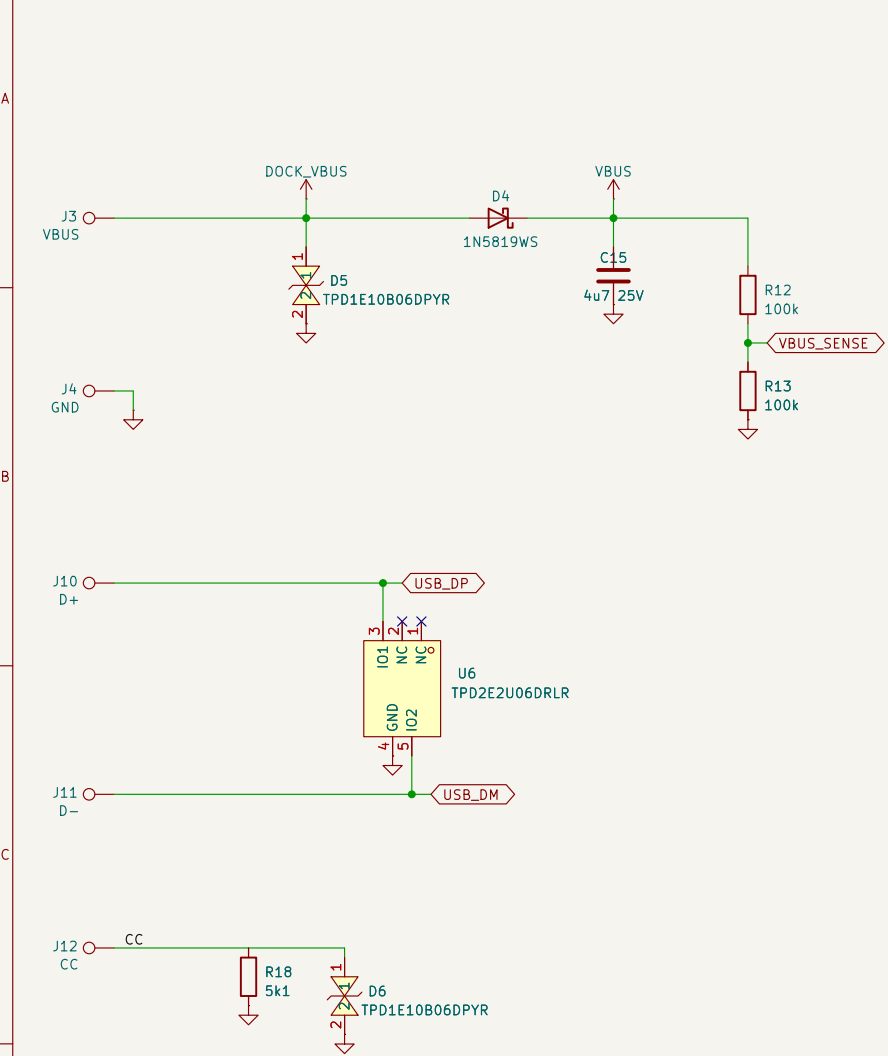
3

4

5

6

Dock contacts and protection



THE DOCK
Five gold pads touch the magnetic charging dock: 5 V power (VBUS), ground, the two USB data lines (D+ / D-) and CC, the line that tells a USB-C charger 'I am a device, send 5 V'.

SPARK GUARD D5
Soaks up static zaps right where they land, on the exposed power contact.

ONE-WAY VALVE D4
Lets dock power in but stops it flowing backwards if the pod is docked the wrong way.

DOCK DETECT R12 / R13
Halves the dock voltage so the brain can safely see 'I am docked'.

USB GUARD U6
Spark protection for the two USB data lines on their way to the brain.

SMOOTHING C15
A small reservoir that steadies the incoming dock power (rated 25 V for margin).

SPARK GUARD D6
The same guard on the CC contact, the last exposed contact that had none (Rev G).

Rail markers (drawing aids for the checker, not parts)

DOCK_VBUS

Plain-language boxes explain each block; part numbers and values are on the symbols.
Generated by hw/pod/sch.py from hw/pod/pod.net (gen.py). Do not edit by hand.

Stereo Ultrasound

Sheet: /1 Dock and USB protection/

File: dock.kicad_sch

Title: Pod board Rev G: Dock and USB protection

Size: A4

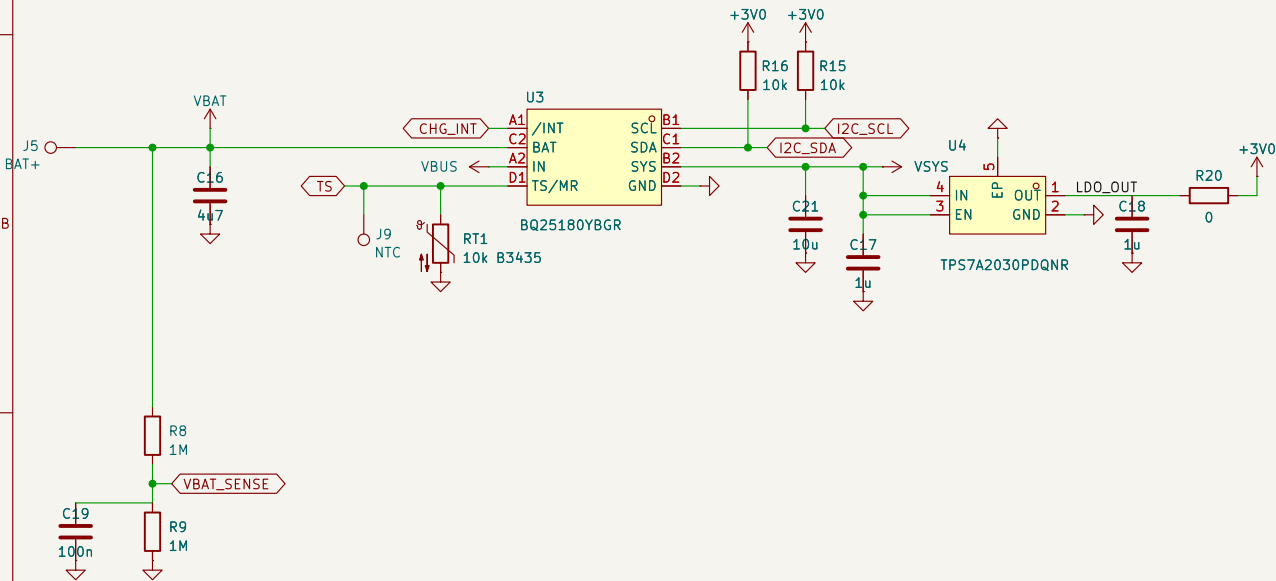
Date: 2026-10-02

Rev: G

KiCad E.D.A. 10.0.6

Id: 2/6

Charger, battery and the 3.0 V supply



Rail markers (drawing aids for the checker, not parts)



BATTERY CHARGER U3
Fills the battery safely, refuses to charge when too hot or cold, and while docked runs the pod straight from the dock. The brain sets it up over two wires (SCL / SDA).

TEMPERATURE CHECK RT1 / J9
A heat-sensitive resistor (or one taped to the cell via J9) tells the charger the battery temperature.

QUIET 3.0 V SUPPLY U4
Turns battery voltage (3.0–4.2 V) into a steady, clean 3.0 V. No switching noise near the microphone.

R20 0 OHM LINK
Cut or lift it to measure the whole pod's current with a meter.

BATTERY GAUGE R8 / R9
Halves the battery voltage so the brain can read how full it is (1 M each: almost no drain).

BATTERY J5
Battery + wire. Battery – shares the ground pad J4 on the dock sheet.

Plain-language boxes explain each block; part numbers and values are on the symbols.
Generated by hw/pod/sch.py from hw/pod/pod.net (gen.py). Do not edit by hand.

Stereo Ultrasound

Sheet: /2 Charger, battery, 3.0 V/

File: power.kicad_sch

Title: Pod board Rev G: Charger, battery, 3.0 V

Size: A4

Date: 2026-10-02

Rev: G

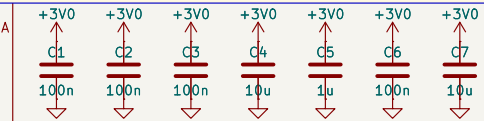
KiCad E.D.A. 10.0.6

Id: 3/6

The brain: STM32U575 microcontroller

DECOUPLING C1–C7

Tiny local energy stores beside the chip's power pins; they keep 3.0 V steady when the chip draws sudden gulps.



THE BRAIN U1

Hears the microphone, shifts ultrasound down into hearing range, drives the speaker, and runs the button, light, battery checks and USB updates. Right-side flags name where each pin goes.

OWN POWER CONVERTER L1, C8, C9

The chip's built-in efficient supply for its core (1.1 V); the only switching supply in the pod.

BUTTON SW1, R10

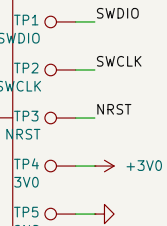
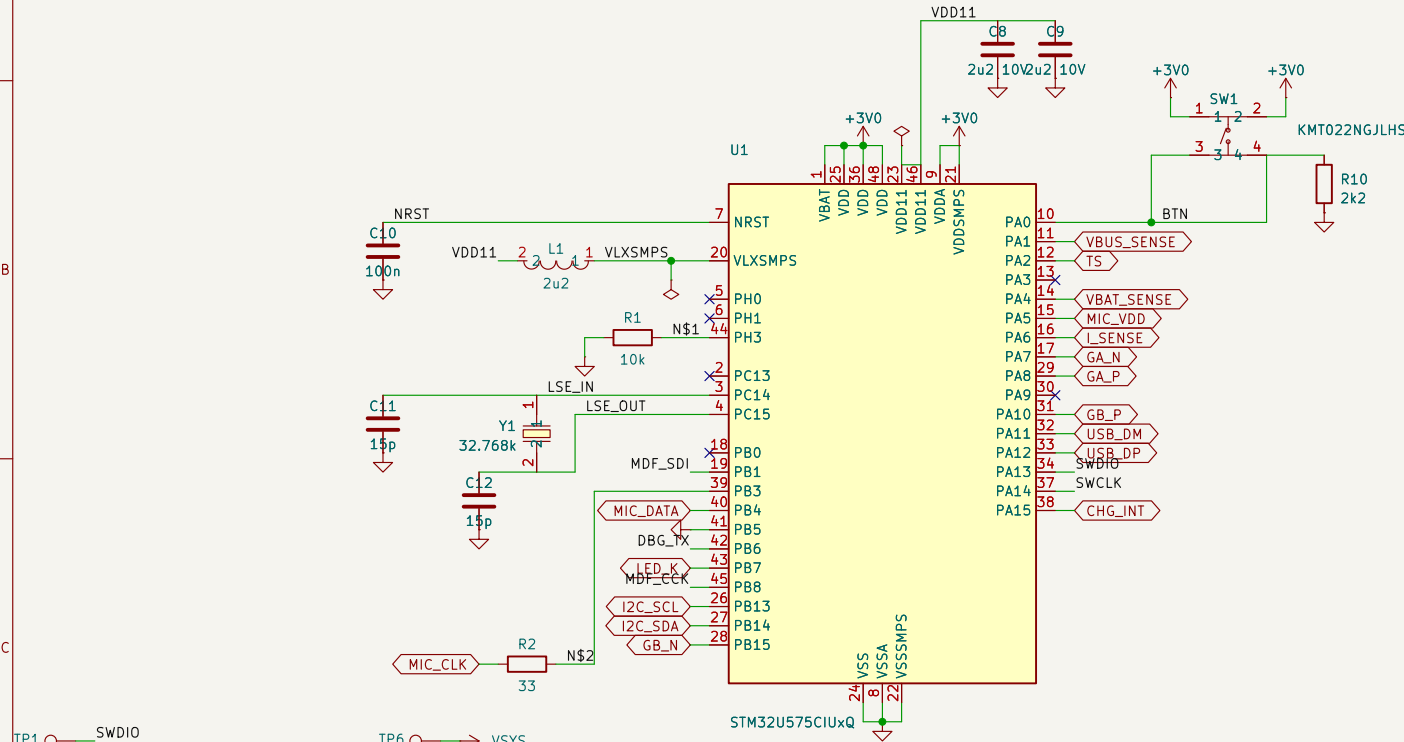
Press = 3.0 V reaches PA0 and wakes the brain; R10 holds it low otherwise.

CLOCK Y1, C11, C12

A 32.768 kHz crystal: the chip's accurate time base.

BOOT AND RESET R1, C10

R1 makes the chip start its own program; C10 keeps the reset line quiet.



TEST PADS TP1–TP10

Gold dots for a programmer (SWDIO, SWCLK, NRST) or a probe. TP7 prints debug text; TP8–TP10 are a hand-wire fallback for the microphone clock.

Plain-language boxes explain each block; part numbers and values are on the symbols. Generated by `hw/pod/sch.py` from `hw/pod/pod.net` (`gen.py`). Do not edit by hand.

Stereo Ultrasound

Sheet: /3 Microcontroller/

File: `mcu.kicad_sch`

Title: Pod board Rev G: Microcontroller

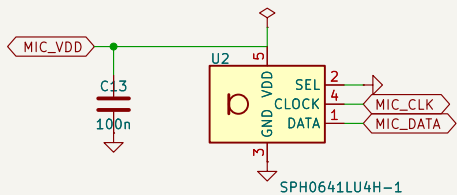
Size: A4 | Date: 2026–10–02

KiCad E.D.A. 10.0.6

Rev: G

Id: 4/6

Ultrasonic microphone



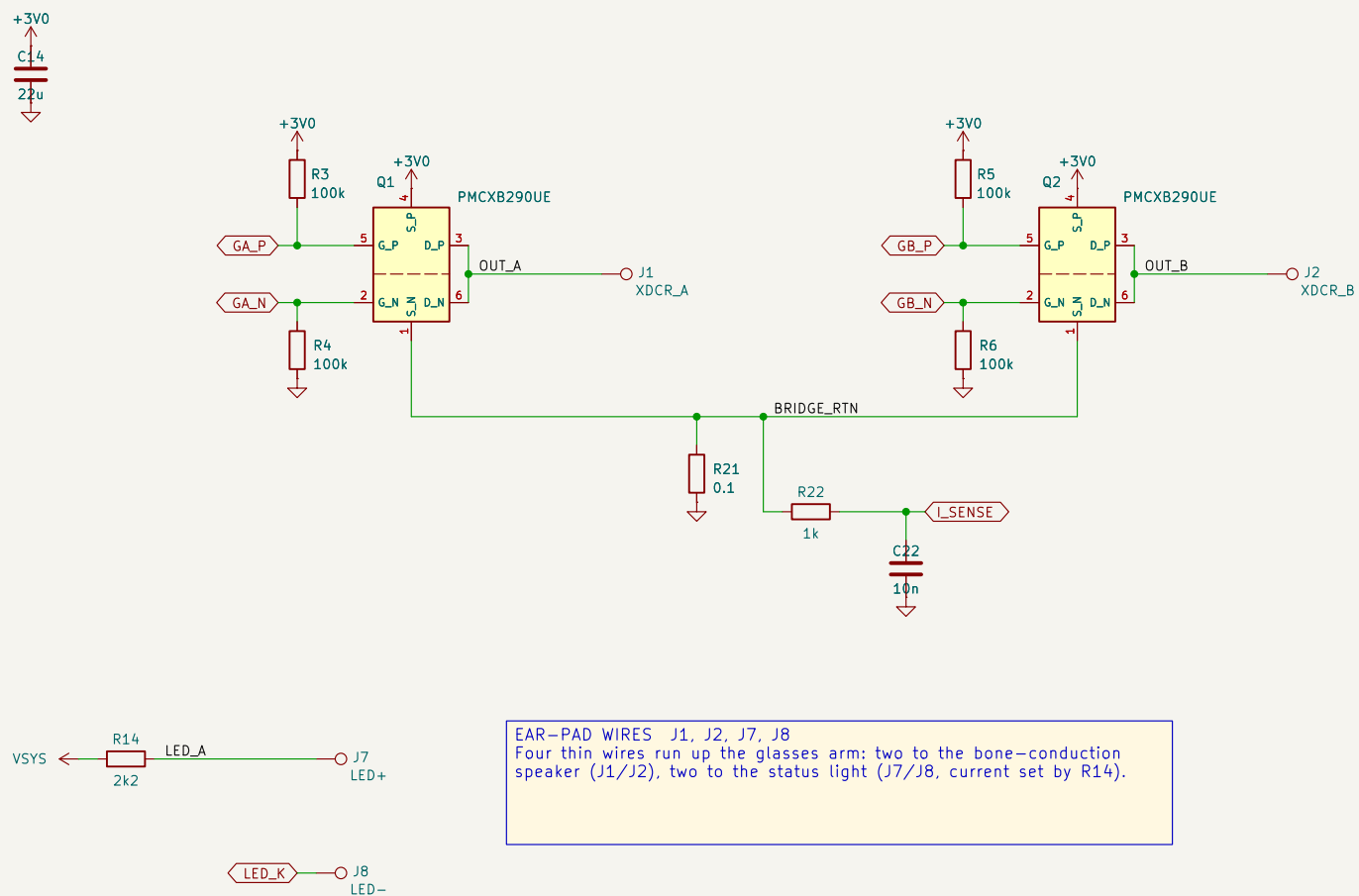
ULTRASONIC MICROPHONE U2
Listens through a 0.6 mm pinhole in the board and a mesh-covered hole in the lid. It sends sound to the brain as a fast stream of 1s and 0s (PDM) timed by MIC_CLK.

POWER SWITCH
The brain powers the mic from a pin (MIC_VDD), so it is fully off when the pod is off. C13 steadies that supply.

SEL TIED TO GROUND
Picks which clock edge the mic answers on.

Plain-language boxes explain each block; part numbers and values are on the symbols. Generated by hw/pod/sch.py from hw/pod/pod.net (gen.py). Do not edit by hand.		
Stereo Ultrasound		
Sheet: /4 Microphone/ File: mic.kicad_sch		
Title: Pod board Rev G: Microphone		
Size: A4	Date: 2026-10-02	Rev: G
KiCad E.D.A. 10.0.6		Id: 5/6

Speaker driver, self-check and ear-pad wires



SPEAKER DRIVER Q1, Q2
Each chip is one P + one N switch. Together they flip the speaker's voltage back and forth 200,000 times a second; the speaker only follows the slow sound part.

SAFE OFF R3-R6
Hold every switch OFF while the brain is starting up or resetting.

SELF-CHECK R21, R22, C22
A 0.1 ohm resistor measures speaker current, so the pod can test its own speaker.

C14 22 uF
Local reservoir for the speaker's current peaks.

EAR-PAD WIRES J1, J2, J7, J8
Four thin wires run up the glasses arm: two to the bone-conduction speaker (J1/J2), two to the status light (J7/J8, current set by R14).